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ABOUT ME

Electronic Engineering Technology graduate specializing in Instrumentation and Control from KMUTNB. Experienced in PLCs, Embedded Systems, Industrial Communication, and Automation through academic and industrial projects. Skilled in ESP32, Raspberry Pi, Modbus TCP, HMI development, and control system design. Interested in Industrial Automation, Robotics, and Smart Agriculture technologies.

EXPERIENCE

Mobile Robot with Bluetooth Connection + Stationary K-LD7 Radar Monitoring

Exchange Program, Hochschule Esslingen (Germany)

APR 2026 – MAY 2026

- Designed a stationary 24 GHz Doppler radar (K-LD7) monitoring system in Python with real-time visualization, safety-zone detection, and CSV logging
- Developed BLE-based mobile robot control (MIT App Inventor + Arduino firmware) for a Makeblock robot, including manual drive and obstacle-avoidance modes.
- Built an autonomous go-to-goal navigation system combining radar position feedback with calibrated motor timing for iterative drive-and-correct movement to target coordinates.

Prior Solution Co., Ltd., Position: R&D IoT for Agriculture (Internship)

DEC 2025 – MAR 2026

- Researched and evaluated IoT technologies for agricultural applications.
- Designed and developed prototype systems to validate assigned R&D concepts.
- Conducted experiments, data collection, and performance analysis to support product development.
- Collaborated with engineers to develop practical solutions for smart agriculture applications.

Improvement and Design of SCARA Robot Control System

JUN 2025 – SEP 2025

- Designed a robotic arm control system using Raspberry Pi and ESP32.
- Applied Inverse Kinematics for coordinate calculation and robot motion control through a GUI.
- Implemented I2C communication between Raspberry Pi (master) and ESP32 (slave).
- Programmed robot operation modes including automatic control, free movement, picking and placing objects within the workspace.

Automatic Wolffia (Duckweed) Farming System Controlled via Application

DEC 2024 – FEB 2025

- Developed an automated smart farming system using PLC, ESP32, and Modbus TCP communication.
- Built a mobile application for remote monitoring and control using Thunkable.
- Integrated Google Sheets cloud database for real-time data exchange.
- Collaborated on PLC programming and HMI development for system operation and monitoring.

EDUCATION

B.Eng. in Electronic Engineering Technology (Instrumentation and Control)

JUL 2022 – MAR 2026

King Mongkut's University of Technology North Bangkok

GPA: 3.45/4.00, Second-Class Honors

SKILLS AND ABILITIES

Technical Skills:

- Hardware/Platforms: ESP32, Arduino (Basic), Raspberry Pi (Basic), PLC - Siemens, Samkoon (Basic)
- Software/Applications: Blynk (Basic), Wonderware InTouch (Basic), Visual Studio Code
- Industrial Communication: Modbus RTU (RS-485), Modbus TCP, I2C, BLE
- CAD and Design Tool :AutoCAD (Basic), Onshape (Basic)
- Programming Languages: C, C++ (Basic), Python (Beginner) ,HTML (Beginner),CSS (Beginner)

Language Skills:

- Thai: Advanced
- English: Intermediate

Additional Skills

- Ability to adapt and take responsibility for assigned project tasks
- Fast learner of new technologies with continuous self-development